Live Market Feed

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Real-time Market Data across exchanges and segments can now be availed on your system via WebSocket. WebSocket provides an efficient means to receive live market data. WebSocket keeps a persistent connection open, allowing the server to push real-time data to your systems.

All Dhan platforms work on these same market feed WebSocket connections that deliver lightning fast market data to you. Do note that this is **tick-by-tick event based data** that is sent over the websocket.

You can establish upto five WebSocket connections per user with 5000 instruments on each connection.

All request messages over WebSocket are in JSON whereas all response messages over WebSocket are in Binary. You will require WebSocket library in any programming language to be able to use Live Market Feed along with Binary converter.

Using DhanHQ Libraries for WebSockets

Establishing Connection

To establish connection with DhanHQ WebSocket for Market Feed, you can connect to the below endpoint using WebSocket library.

```
wss://api-feed.dhan.co?version=2&token=<your-access-token>&clientId=<your-client-id>&authType=2
```

Query Parameters

FIELD	DESCRIPTION
version*	2 for DhanHQ v2
token*	Access Token generated via Dhan
clientId*	User specific identification generated by Dhan
authType*	2 by Default

Adding Instruments

You can subscribe upto 5000 instruments in a single connection and receive market data packets. For subscribing, this can be done using JSON message which needs to be send over WebSocket connection.

Note

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You can only send upto 100 instruments in a single JSON message. You can send multiple messages over a single connection to subscribe to all instruments and receive data.

Request Structure

```
{
  "RequestCode" : 15,
  "InstrumentCount" : 2,
  "InstrumentList" : [
    {
      "ExchangeSegment" : "NSE_EQ",
      "SecurityId" : "1333"
    },
    {
      "ExchangeSegment" : "BSE_EQ",
      "SecurityId" : "532540"
    }
  ]
}
```

Parameters

FIELD (REQUIRED*)	DESCRIPTION
RequestCode * int	Code for subscribing to particular data mode (see Annexure)
InstrumentCount * int	No. of instruments to subscribe from this request
InstrumentList.ExchangeSegment * enum string	Exchange Segment of instrument to be subscribed (see Annexure)
InstrumentList.SecurityId * string	Exchange standard ID for each scrip. refer here

Keeping Connection Alive

To keep the WebSocket connection alive and prevent it from closing, the server side uses **Ping-Pong** module. Server side sends ping every 10 seconds to the client server (in this case, your system) to maintain WebSocket status as open.

An automated pong is sent by websocket library. You can use the same as response to the ping.

warning

In case the client server does not respond for more than 40 seconds, the connection is closed from server side and you will have to reestablish connection.

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Market Data

The market feed data is sent as structured binary packet which is shared at super fast speed.

DhanHQ Live Market Feed is real-time data and there are three modes in which you can receive the data, depending on your use case:

- Ticker Data
- Quote Data
- Full Data

Binary Response

Binary messages consist of sequences of bytes that represent the data. This contrasts with text messages, which use character encoding (e.g., UTF-8) to represent data in a readable format. Binary messages require parsing to extract the relevant information.

The reason for us to choose binary messages over text or JSON is to have compactness, speed and flexibility on data to be shared at lightning fast speed.

All responses from Dhan Market Feed consists of Response Header and Payload. Header for every response message remains the same with different feed response code, while the payload can be different.

Endianness Endianness determines the order in which bytes are arranged for multi-byte data (like integers and floats).

Types:

- **Little Endian:** Least significant byte first (0x78, 0x56, 0x34, 0x12)
- **Big Endian:** Most significant byte first (0x12, 0x34, 0x56, 0x78)

The data on DhanHQ Websockets are sent in Little Endian. In case your system is Big Endian, you will have to define endianness while reading the websocket.

Response Header

The response header message is of 8 bytes which will remain same as part of all the response messages. The message structure is given as below.

BYTES	TYPE	SIZE	DESCRIPTION
0	[ ] byte	1	Feed Response Code can be referred in Annexure
1-2	int16	2	Message Length of the entire payload packet
3	[ ] byte	1	Exchange Segment can be referred in Annexure

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BYTES	TYPE	SIZE	DESCRIPTION
4-7	int32	4	Security ID - can be found here

Ticker Packet

This packet consists of Last Traded Price (LTP) and Last Traded Time (LTT) data across segments.

BYTES	TYPE	SIZE	DESCRIPTION
0-7	[ ] array	8	Response Header with code `2` (see Annexure)
8-11	float32	4	Last Traded Price of the subscribed instrument
12-15	int32	4	Last Trade Time (EPOCH)

Prev Close Whenever any instrument is subscribed for any data packet, we also send this packet which has Previous Day data to make it easier for day on day comparison.

BYTES	TYPE	SIZE	DESCRIPTION
0-7	[ ] array	8	Response Header with code `6` (see Annexure)
8-11	float32	4	Previous day closing price
12-15	int32	4	Open Interest - previous day

Quote Packet

This data packet is for all instruments across segments and exchanges which consists of complete trade data, along with Last Trade Price (LTP) and other information like update time and quantity.

BYTES	TYPE	SIZE	DESCRIPTION
0-7	[ ] array	8	Response Header with code `4` (see Annexure)
8-11	float32	4	Latest Traded Price of the subscribed instrument

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BYTES	TYPE	SIZE	DESCRIPTION
12-13	int16	2	Last Traded Quantity
14-17	int32	4	Last Trade Time (LTT) - EPOCH
18-21	float32	4	Average Trade Price (ATP)
22-25	int32	4	Volume
26-29	int32	4	Total Sell Quantity
30-33	int32	4	Total Buy Quantity
34-37	float32	4	Day Open Value
38-41	float32	4	Day Close Value - only sent post market close
42-45	float32	4	Day High Value
46-49	float32	4	Day Low Value

OI Data Whenever you subscribe to Quote Data, you also receive Open Interest (OI) data packets which is important for Derivative Contracts.

BYTES	TYPE	SIZE	DESCRIPTION
0-7	[ ] array	8	Response Header with code `5` (see Annexure)
8-11	int32	4	Open Interest of the contract

Full Packet

This data packet is for all instruments across segments and exchanges which consists of complete trade data along with Market Depth and OI data in a single packet.

BYTES	TYPE	SIZE	DESCRIPTION
0-7	[ ] array	8	Response Header with code `8` (see Annexure)
8-11	float32	4	Latest Traded Price of the subscribed

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BYTES	TYPE	SIZE	DESCRIPTION
			instrument
12-13	int16	2	Last Traded Quantity
14-17	int32	4	Last Trade Time (LTT) - EPOCH
18-21	float32	4	Average Trade Price (ATP)
22-25	int32	4	Volume
26-29	int32	4	Total Sell Quantity
30-33	int32	4	Total Buy Quantity
34-37	int32	4	Open Interest in the contract (for Derivatives)
38-41	int32	4	Highest Open Interest for the da (only for NSE_FNO)
42-45	int32	4	Lowest Open Interest for the day (only for NSE_FNO)
46-49	float32	4	Day Open Value
50-53	float32	4	Day Close Value - only sent post market close
54-57	float32	4	Day High Value
58-61	float32	4	Day Low Value
62-161	Market Depth Structure	100	5 packets of 20 bytes each for each instrument in below provided structure

Each of these 5 packets will be received in the following packet structure:

BYTES	TYPE	SIZE	DESCRIPTION
0-3	int32	4	Bid Quantity

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BYTES	TYPE	SIZE	DESCRIPTION
4-7	int32	4	Ask Quantity
8-9	int16	2	No. of Bid Orders
10-11	int16	2	No. of Ask Orders
12-15	float32	4	Bid Price
16-19	float32	4	Ask Price

Feed Disconnect

If you want to disconnect WebSocket, you can send below JSON request message via the connection.

```
{
  "RequestCode" : 12
}
```

In case of WebSocket disconnection from server side, you will receive disconnection packet, which will have disconnection reason code.

- If more than 5 websockets are established, then the first socket will be disconnected with `805` with every additional connection.

BYTES	TYPE	SIZE	DESCRIPTION
0-7	[ ] array	8	Response Header with code `50` (see Annexure)
8-9	int16	2	Disconnection message code - here

You can find detailed Disconnection message code description here.

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DhanHQ is our commitment to empowering India's trading ecosystem with advanced, low-latency infrastructure. Built on the same robust technology that powers Dhan, DhanHQ offers free trading APIs, real-time market data, and a comprehensive suite of tools tailored for traders, developers, and fintech platforms. Whether you're building custom trading systems or embedding wealth products into your product, DhanHQ provides the flexibility and speed you need. With state-of-the-art infrastructure and a developer-friendly approach, we're enabling a new era of trading innovation in India.

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SEBI Stock Broker Registration No: INZ000006031 | Depository Participant (CDSL) ID: IN-DP-289-2016
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